

Mammography Model Development Recommendation

Recommended next-stage research plan after the current v12 ensemble and preprocessing experiments

Recommendation: reopen model development. Freeze the evaluation protocol and the validated preprocessing choices, but do not treat v12 as the final model. The strongest next step is a target-specific, high-resolution, lesion-aware, multi-view screening model trained directly for breast cancer detection.

1. Why the current approach has reached its limit

- **Training task mismatch:** the current model was trained on a three-class task (Normal / Benign / Malignant), while the main external screening target is cancer versus no cancer.
- **Domain mismatch:** the model is trained on curated multi-source mammography datasets but is being evaluated on raw RSNA screening mammograms.
- **Resolution limitation:** the current pipeline uses approximately 384-512 px inputs. Small malignant findings, especially calcifications, can disappear under aggressive downscaling.
- **Fusion limitation:** averaging or maximising image-level probabilities does not test true learned multi-view reasoning.
- **Source shortcut risk:** class prevalence differs strongly across datasets, so a model can learn dataset identity and source-specific priors instead of pathology.

Small preprocessing changes can still be useful for stability, but they are unlikely to close the performance gap on their own. The architecture, target, resolution and training data strategy now matter more.

2. First action: audit the RSNA data coverage

Before training a new model, verify why the current RSNA manifest contains substantially fewer images than the full official release. This audit should distinguish deliberate exclusions from decoding, metadata, file-mapping or filtering failures.

- Compare the current manifest against the original RSNA train.csv.
- Identify every excluded patient and image.
- Count L-CC, R-CC, L-MLO and R-MLO images per examination.
- Check whether missing views come from real acquisition differences or pipeline errors.
- Check duplicates across RSNA and Mammo-Bench sources using IDs and perceptual hashes.
- Document all exclusions so the final evaluation is reproducible.

3. Recommended model architecture

Build a shared high-resolution CNN encoder with a global image branch, a local suspicious-region branch and a learned cross-view fusion module.

Component	Recommended design
Input	1536x768 or 2048x1024 mammograms
Encoder	ConvNeXt-Small or ConvNeXt-Base with shared weights
Global branch	Whole-breast representation
Local branch	Top suspicious high-resolution patches
View encoding	CC/MLO and left/right embeddings
Fusion	Cross-view transformer or attention module
Robustness	Missing-view masks and bilateral comparison
Primary head	Breast-level cancer probability
Auxiliary heads	Normal/Benign/Malignant and optional lesion localisation

Why this is different from the previous fusion experiment

The previous experiment combined completed image probabilities. The proposed model learns relationships between CC and MLO views before the final prediction. That allows the network to compare morphology, position, asymmetry and suspicious regions across views.

4. Research programme in priority order

Priority	Experiment	Purpose
0	Audit RSNA coverage, labels and duplicates	Confirm that images were not lost accidentally
1	High-resolution ConvNeXt trained directly on RSNA cancer labels	Correct the main task mismatch
2	Add compatible external mammography data	Increase cancer cases and scanner diversity
3	Global-local patch learning	Preserve small lesions and improve localisation
4	Learned four-view and bilateral fusion	Use information that score averaging cannot learn
5	Mammography-specific pretraining	Test stronger transfer representations
6	Domain-robust training	Reduce source and scanner shortcut learning
7	Ensemble only genuinely different models	Combine complementary error patterns

5. First serious baseline

The next reproducible baseline should follow the design principles of successful RSNA systems, while using a clean patient-separated split and training from scratch on the project data.

- Correct DICOM decoding.
- Frozen 0.1-99.9 windowing.
- Breast ROI crop without pectoral removal.
- ConvNeXt-Small backbone.
- 2048x1024 input resolution.
- Binary breast-cancer target.
- Four patient-separated folds.
- Compare weighted BCE and focal loss.
- Use exponential moving average where appropriate.
- Store out-of-fold breast-level predictions for calibration and error analysis.

6. Lesion-aware branch

After the high-resolution baseline is stable, add a weakly supervised global-local branch. The global network proposes suspicious regions; the local network analyses selected patches at higher effective resolution. If lesion annotations from datasets such as VinDr, INbreast or CBIS-DDSM are available, use them for detector or localisation pretraining. Otherwise, use weakly supervised top-patch selection.

7. Source-shortcut controls

- Balanced cancer/non-cancer sampling within each source.
- Uniform raw-image preprocessing across datasets.
- Random gamma, contrast, noise and resolution augmentation.
- Multi-view contrastive learning using CC and MLO views from the same breast.
- Source-adversarial training only if a source-probe remains strong.
- Leave-one-source-out validation.
- GroupDRO only as a candidate objective, because previous testing reduced malignant recall.
- Do not provide source, scanner or site identifiers as prediction features.

8. Suggested success targets

Measure	Aspirational target
Sealed RSNA holdout AUROC	≥ 0.80
Sensitivity at ~3.5% recall	$\geq 60\%$
Calibration ECE	≤ 0.03
Major subgroups	No adequately powered subgroup below chance
Mammo-Bench patient-level macro F1	≥ 0.75
Mammo-Bench malignant recall	≥ 0.80

9. Immediate implementation sequence

- **Step 1:** Audit the RSNA manifest and recover any unintentionally excluded images.
- **Step 2:** Build a clean high-resolution RSNA binary baseline with ConvNeXt-Small.
- **Step 3:** Establish patient-level out-of-fold predictions and sealed external evaluation.
- **Step 4:** Add the global-local lesion branch.
- **Step 5:** Add learned feature-level multi-view fusion.
- **Step 6:** Test mammography-specific pretraining and domain-robust objectives.
- **Step 7:** Ensemble only after individual models are independently competitive.

Highest-priority recommendation: do not run another small preprocessing experiment first. Audit the missing RSNA images, then train a 2048x1024 ConvNeXt cancer-detection baseline.

10. Materials needed for the next stage

- Complete RSNA manifest and original train.csv.
- Current preprocessing and model-training notebooks.
- Patient-level split files.
- Current checkpoints and out-of-fold predictions.
- Original RSNA DICOM files.
- Optional: VinDr images/annotations, EMBED public cohort or other permitted mammography datasets.

Selected references

- RSNA Breast Cancer Detection winning implementation: https://github.com/dangnh0611/kaggle_rsna_breast_cancer
- GMIC implementation: <https://github.com/nyukat/gmic>
- GLAM study: <https://pmc.ncbi.nlm.nih.gov/articles/PMC8791642/>
- Mammo-FM v2 preprint: <https://arxiv.org/abs/2512.00198>
- Mammo-FM repository / weights licence: <https://github.com/batmanlab/Mammo-FM>
- Mammo-Bench repository: <https://github.com/Gaurav2543/Mammo-Bench>

Prepared as a practical development roadmap for the capstone mammography classification project.